

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

```
df=pd.read_csv("iris.csv")
```

```
df.head(150)
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

```
[150 rows x 5 columns]
```

```
df.shape
```

```
(150, 5)
```

```
df.loc[df['variety']=='Iris-virginica']
```

```
Empty DataFrame
```

```
Columns: [sepal.length, sepal.width, petal.length, petal.width, variety]
```

```
Index: []
```

```
df_setosa=df.loc[df['variety']=='setosa']
```

```
df_virginica=df.loc[df['variety']=='Virginica']
```

```
df_versicolor=df.loc[df['variety']=='Versicolor']
```

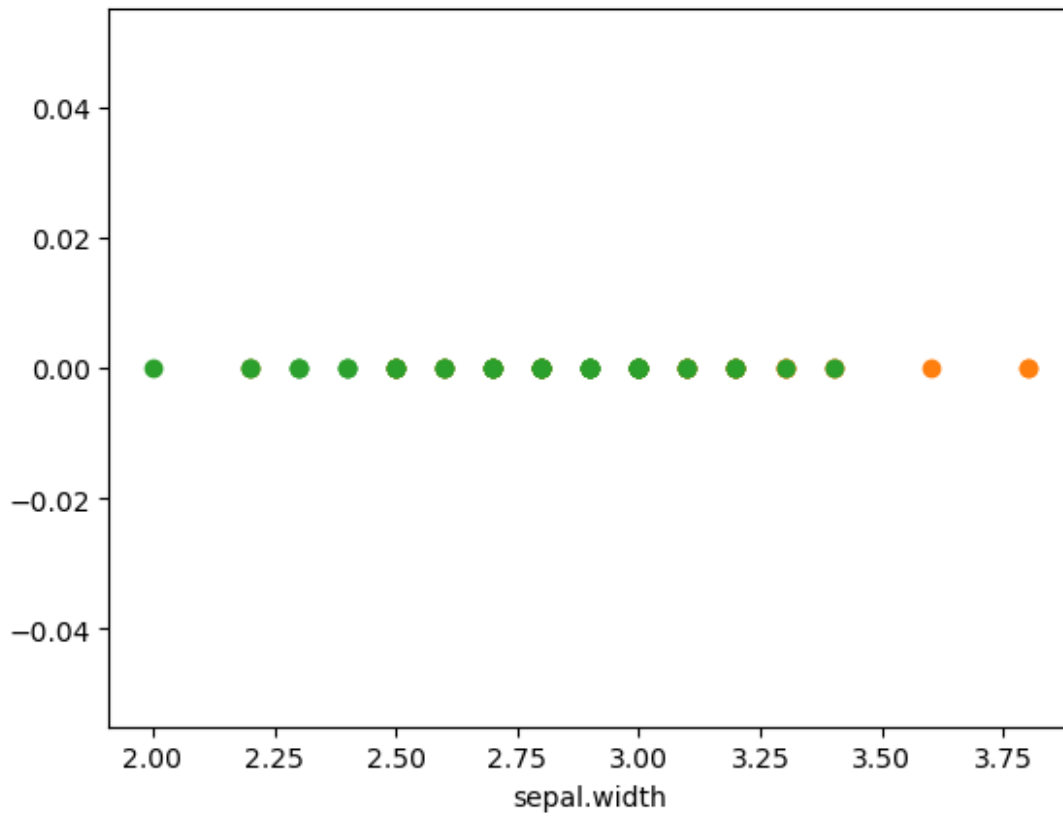
```
plt.scatter(df_setosa['sepal.width'],np.zeros_like(df_setosa['sepal.width']))
```

```
plt.scatter(df_virginica['sepal.width'],np.zeros_like(df_virginica['sepal.width']))
```

```
plt.scatter(df_versicolor['sepal.width'],np.zeros_like(df_versicolor['sepal.width']))
```

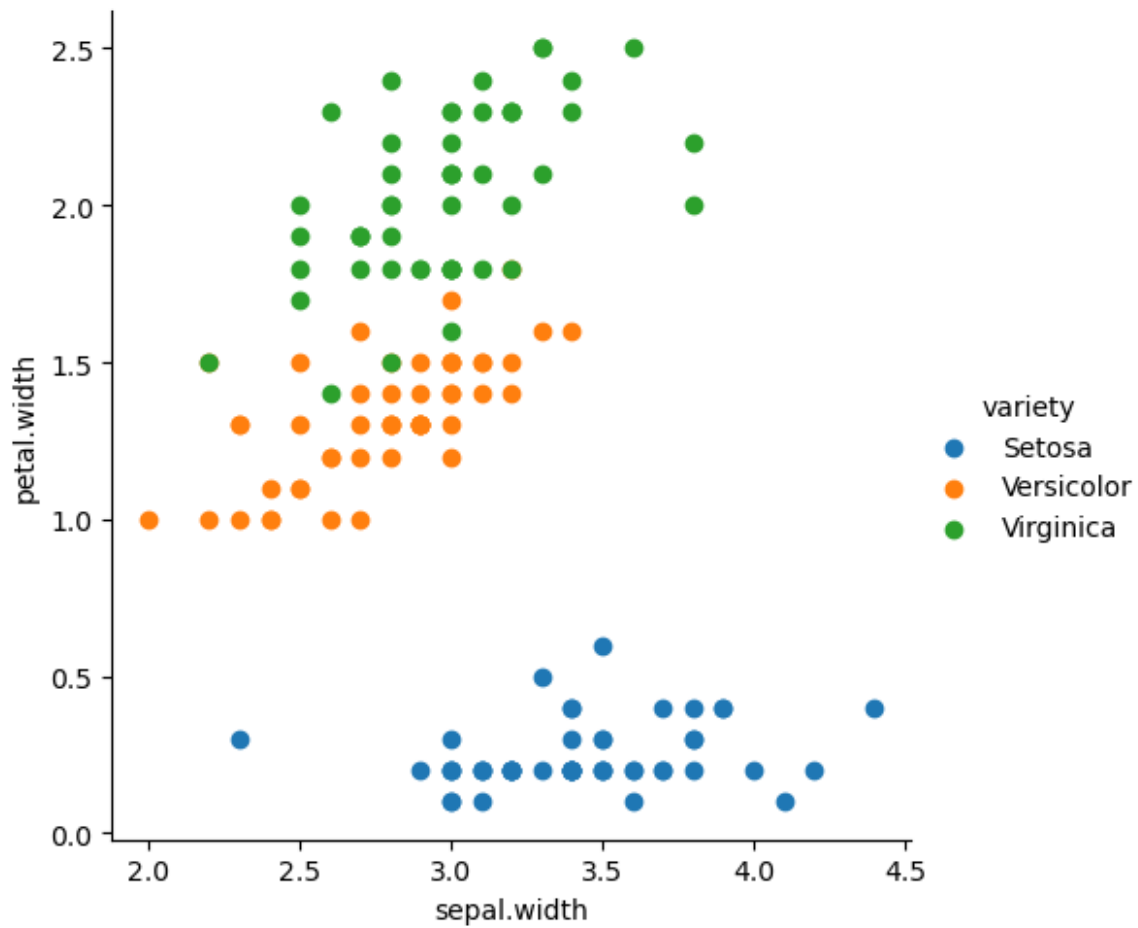
```
plt.xlabel('sepal.width')
```

```
plt.show()
```

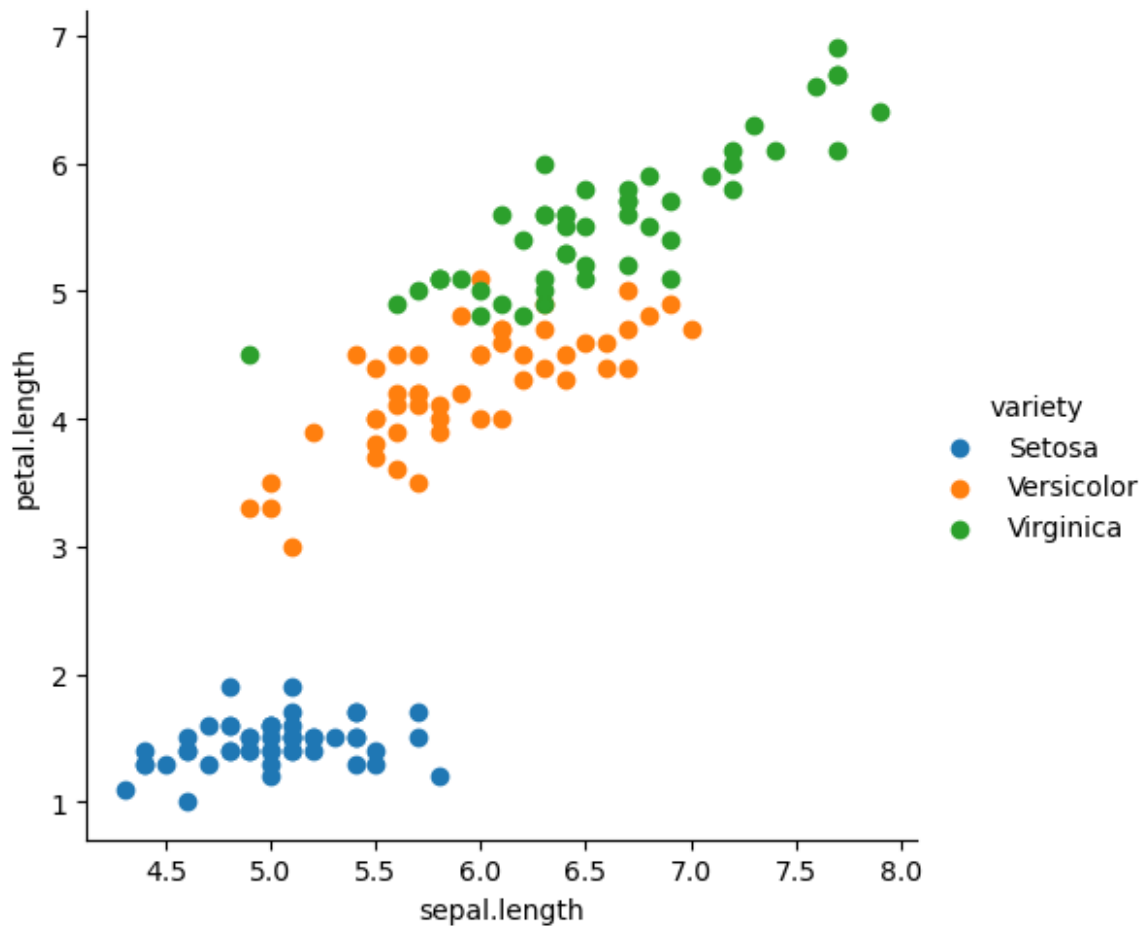


```
sns.FacetGrid(df, hue = 'variety', height =  
5).map(plt.scatter, "sepal.width", "petal.width").add_legend()
```

```
<seaborn.axisgrid.FacetGrid at 0x2514dd8d000>
```



```
sns.FacetGrid(df,hue = 'variety', height =  
5).map(plt.scatter,"sepal.length","petal.length").add_legend()  
<seaborn.axisgrid.FacetGrid at 0x25149f9f490>
```



```
sns.pairplot(df,hue="variety",size =2)
```

```
C:\Users\DELL\anaconda3\lib\site-packages\seaborn\axisgrid.py:2095:  
UserWarning: The `size` parameter has been renamed to `height`; please  
update your code.
```

```
warnings.warn(msg, UserWarning)
```

```
<seaborn.axisgrid.PairGrid at 0x25149e03b20>
```

